

SOUMYASREE SABAT

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TECHNICAL SUMMARY

Backend-focused software engineer with experience building scalable systems, security-aware middleware, blockchain infrastructure, and data-driven applications. Strong foundation in distributed systems, optimization, and backend architecture with hands-on experience across Python, C++, cloud tooling, and secure API design. Passionate about engineering resilient systems with measurable performance and reliability improvements.

EDUCATION

B.Tech, Computer Science & Engineering

Oct 2023 – Present

National Institute of Science and Technology (NIST), Berhampur, Odisha | **CGPA: 9.34/10.**

ISC Class 12 — 91% | ICSE Class 10 — 91.5%

De Paul School, Berhampur

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, Go, Java, SQL, Solidity, Bash.

Frameworks & Libraries: Flask, Django, Next.js, Pandas, EtherJS, HardHat, Metasploit.

Tools & Platforms: Git, AWS, Docker, Kubernetes, Linux, Tableau.

Domains: Backend Engineering, Distributed Systems, Blockchain/Web3, ML/Anomaly Detection, OffSec.

EXPERIENCE

Software Development Intern · Protionix Technologies

May 2026 – Ongoing

- Diagnosed and resolved software defects through root-cause analysis and systematic debugging workflows, that improved optional stability.
- Engineered backend-integrated web platforms with scalable architecture patterns to minimize system bottlenecks and improve maintainability.
- Teamed up in peer code reviews with a 3-person engineering team, improving codebase reliability and accelerating fix deployment cycles.

Data Analytics Intern · NIST University

May 2025 – Jun 2025

- Executed 8+ real-world case studies using Python (Pandas, NumPy, Matplotlib), translating raw institutional datasets into actionable operational insights.
- Built a workforce scheduling optimization model that reduced scheduling conflicts and delays by **30%** across university departments.
- Delivered full-cycle analytics pipeline: CSV ingestion → data cleaning → visualization → executive-ready insight reports.

Data Structures & Algorithms Intern · NIST University

Jul 2024 – Aug 2024

- Engineered 5+ optimized solutions in C targeting sub- $O(n^2)$ complexity for algorithmic problem sets under timed competitive conditions.
- Reduced average runtime of target programs by ~30% through application of efficient data structures (heaps, tries, segment trees).

C++ /C Intern · Robogenesis

May 2024 – Jul 2024

- Implemented move-generation, and state-management logic for turn-based game environments.
- Leveraged foundational game theory strategic concepts to improve the logic-based decision making

PROJECTS

AI Detect — LLM Security Middleware | Python, Flask, Sandbox Execution

Ongoing

- Engineered a request/response interceptor layer for LLM-backed APIs performing input validation, prompt injection detection, and sensitive data screening on every transaction.
- Implemented sandboxed execution environment to contain adversarial payloads before surfacing model responses and reduce exposure to prompt poisoning and unsafe model behaviour.

HeirChain — Blockchain Document Verification Platform | Solidity, EtherJS, Node.js *Ongoing*

- Architected a decentralized multi-staged document verification on Ethereum using Solidity smart contracts and cryptographic digital signature validations.
- Engineered tamper-resistant document validation using Solidity smart contracts with immutable on-chain audit trails for hierarchical approval systems.
- Designed asynchronous notification layer and configurable approval-routing engine supporting multi-tier authority hierarchies and automated verification flows.

HourGlass — AI-Based Academic Scheduler | Python, Constraint Optimization *July 2026*

- Engineered constraint-based timetable generation modeling instructor availability, room allocation and scheduling conflicts.
- Automated scheduling workflows using optimization-based allocation logic, eliminating manual conflict.

CodeLens — RAG-Based Codebase Intelligence Tool | Python, NextJS, RAG *April 2026*

- Built a retrieval-augmented generation system over code repositories, letting developers query functions, logic flows, and documentation in plain language, cutting context-switching time during debugging.
- Designed a chunking and embedding pipeline tailored for code structure, preserving function scope, class hierarchies, and file relationships that generic document loaders collapse and lose.
- Integrated a local LLM backend to keep sensitive codebases off third-party APIs, with response latency optimized under real query loads.

Invoice Risk Detection — ML Anomaly Pipeline | Python, Scikit-learn, Pandas *Dec 2025*

- Developed supervised classification model to flag high-risk vendor invoices, analyzing anomalies in freight costs, pricing patterns, and delivery metrics across transaction datasets.
- Automated risk-scoring pipeline to prioritize suspicious transactions, cutting unnecessary manual audit reviews and improving financial auditing efficiency.

SnapDrop — Secure P2P File Transfer Platform | Python, Next.js *July 2025*

- Built a cross-device peer-to-peer file sharing platform with AES-secured transfer channels and token-based authentication access control.
- Designed cloud-backed distribution storage architecture with configurable public/private visibility, and handling uploads.
- Streamlined transfer pipeline and frontend interaction workflow to achieve near real-time responsiveness.

ACHIEVEMENTS & ACTIVITIES

- **Bharatiya Antariksh Hackathon (ISRO)** — Selected nationally to compete in the Indian Space Research Organisation's space technology hackathon.
- **Advent of Cyber 2025 (TryHackMe)** — Completed 24-day hands-on cybersecurity challenge series covering network exploitation, threat analysis, and forensics.
- **Linux Foundation Certification** — System administration, CLI tooling, and process management.